

Dissertation

Delsad, Sebastien, DS
Arnold, Melvin, MX
Logean, Romain, SIQ

Study of the conditions for the emergence of altruism through computer simulations

Professor: Christian Sachse
Philosophy of Life Sciences
Submission date: 04.06.2025

Lausanne, academic year 2024–2025

English translation of the original French document, produced with the assistance of a LLM.

The logo of EPFL (École Polytechnique Fédérale de Lausanne) is displayed in a bold, red, sans-serif font. The letters are stylized, with the 'E' and 'F' having a distinctive blocky appearance.

Contents

1	Introduction	1
2	Theoretical framework [1]	1
2.1	Biological altruism	1
2.2	Behavioral altruism	2
2.3	Psychological altruism	3
2.4	Links between definitions	3
2.5	Link to game theory	4
2.6	Related work	5
3	Methodology: Computer simulation of altruism	6
3.1	Presentation of the simulation model	6
3.1.1	Definition of the environment	6
3.1.2	Introduction of agents	7
3.2	Simulation parameters and learning of the autonomous agent	8
3.3	Assumptions and limitations of the model	8
4	Results	10
4.1	Problems encountered	10
4.2	Intelligent agent vs semi-random	10
4.2.1	Intelligent agent vs giving	11
4.2.2	Intelligent agent vs attacking	11
4.3	Intelligent agent vs random	12
4.4	Semi-random vs random	12
4.4.1	Giving vs random	12
4.4.2	Attacking vs random	13
4.4.3	Complete vs random	13
4.5	Semi-random vs semi-random	13
4.5.1	Complete vs giving vs attacking	13
5	Discussion: Relevance and implications of the results	14
5.1	Analysis of the results	14
5.1.1	Varied parameters	14
5.1.2	Intelligent agent vs semi-random	14
5.1.3	Semi-random vs random	16
5.1.4	Semi-random vs semi-random	17
6	Conclusion	17

1 Introduction

Altruism is a complex notion that intrigues biologists as much as psychologists and philosophers. In her book *Je t'aide... moi non plus*, Christine Clavier proposes three distinct definitions of altruism: biological, behavioral, and psychological. These definitions reflect different but complementary perspectives, each belonging to a specific field. Biological altruism focuses on the impact of behaviors on the fitness (reproductive success) of individuals, behavioral altruism is defined by observable acts aimed at benefiting others, while psychological altruism emphasizes the internal intentions and motivations of individuals. According to the author, despite their differences, these three notions of altruism intersect without overlapping. Indeed, biological altruism could well be a necessary condition for the evolution of psychological altruism, meaning that these notions are related without being identical.

These three forms of altruism, seemingly distinct, do they not converge toward the same underlying reality? This question is fundamental for better understanding altruistic behaviors as they manifest in nature and in humans. If these three definitions share a common coherence, this could unify still fragmented debates between disciplines by linking evolutionary biology, behavioral sciences, and psychology.

Historically, the question of altruism is rooted in a debate going back to Darwin's work, for whom cooperation posed a paradox to the theory of natural selection. This tension gave rise to theories such as kin selection (Hamilton), reciprocal altruism (Trivers), and more recently game theory models that explore the evolutionary stability of altruistic behaviors. Indeed, to evaluate the emergence of cooperation and the concept of an evolutionarily stable strategy, theorists also use game theory to model interactions between selfish and cooperative individuals. Thus, the use of computer simulations to test a specific example makes it possible to go beyond the limits of empirical observation and to shed new light on the evolutionary and cognitive dynamics of altruism.

Can altruistic behaviors exist? If so, in what environments can they emerge? These are the main questions that will be addressed in this study through computer simulations, with the aim of discussing the different definitions of altruism.

First, the theoretical framework is presented; the different definitions of altruism are discussed. Next, the methodology used for the computer simulation is explained, before presenting the results obtained. Finally, the relevance of the results is discussed and the methodology is critiqued, in order to address the main research question of this work.

2 Theoretical framework [1]

2.1 Biological altruism

The paradox of biological altruism and the emergence of cooperation represent a fundamental challenge in evolutionary biology. Altruism is defined as a behavior that increases the fitness of another individual while reducing that of the individual performing the behavior. At first glance, such a definition appears incompatible with the principles of natural selection. Indeed, if a behavior harms the fitness of the altruistic individual, it seems unlikely that natural selection would favor the transmission of the genetic bases underlying this behavior. Thus, such a behavior should not be preserved over generations. Yet, in the living world, altruistic behaviors are frequently observed, making this an empirical paradox.

To better understand this paradox, one can consider a population composed of altruistic and non-altruistic individuals. In this case, the fitness of altruists is necessarily lower than that of non-altruists, since the latter benefit from altruistic actions without paying the costs. Consequently, the

proportion of non-altruists increases with each generation compared to that of altruists. Iterating this process over several generations leads to the complete disappearance of altruistic individuals. This logic seems to imply that altruism cannot be maintained in a population.

However, several theories have been developed to resolve this paradox by re-evaluating the concepts of selection, fitness, and cooperation. Among them, the theory of reciprocal altruism, proposed by Robert Trivers, plays a central role. This theory postulates that an individual may adopt an altruistic behavior toward another if the latter is likely to return the favor in the future, thereby ensuring mutual long-term benefit. Although this conceptual framework relies on repeated interactions, it helps explain the persistence of certain altruistic behaviors in contexts where reciprocity is possible.

The theory of reciprocal altruism also gave rise to major developments in evolutionary biology, notably the application of game theory to evolution. By modeling social interactions as strategic games, this approach allows the exploration of conditions under which cooperation and altruism can emerge and stabilize within populations. These theories enrich our understanding of evolutionary mechanisms and show that, far from being an insurmountable paradox, altruism can fit naturally within the framework of natural selection, particularly through complex dynamics of cooperation.

2.2 Behavioral altruism

The neoclassical economic model rests on the figure of the *homo economicus*, a rational agent endowed with personal desires and goals, who in all circumstances seeks to maximize profit and personal well-being according to individual preferences. This model assumes that economic agents are always capable of making the most optimal decisions possible. However, it has been the subject of numerous criticisms. Some authors, such as Gerd Gigerenzer, have shown that in many cases, following intuition and emotions proves, on average, more advantageous than systematically acting in a rational manner.

In response to these limitations, researchers opposed to neoclassical economic theory founded experimental economics, an approach that uses methods from experimental psychology to empirically test the validity of existing economic theories. This discipline has made it possible to define a specific form of altruism, referred to as behavioral altruism. According to this view, an action is called altruistic if it involves a cost to the agent performing it, provides a benefit to others, and does not allow the agent to expect any return on investment, whether immediate or delayed. The “cost” often refers to losses of resources such as money or time.

Behavioral altruism differs from biological altruism, so that certain behaviors judged altruistic by biologists are not so according to experimental economists, and vice versa. To study these behaviors, experimental economists rely on experiments in the form of games, such as the *Trust Game* or the *Dictator Game*. However, these experiments are conducted over limited periods of time, which prevents accurately measuring the potentially beneficial effects of certain altruistic actions in the long term. This limitation can restrict the overall understanding of the phenomenon.

A notable manifestation of behavioral altruism revealed during these experiments is that of altruistic punishment. It has been observed that participants, when given the opportunity, punish opportunistic players, even at personal cost. Empirically, these punishments also have a behavioral effect; opportunistic individuals tend to become more altruistic after being sanctioned. Moreover, the mere fact that their actions are observed can influence player behavior.

Contrary to the assumptions of neoclassical economists, altruistic behaviors, though sometimes costly in the short term, often prove optimal in the long term. Furthermore, they can be beneficial at levels beyond that of the individual, notably for groups or even within a genetic framework, thereby reinforcing the importance of altruism in broader contexts.

2.3 Psychological altruism

The notion of psychological altruism lies at the heart of an old and still relevant philosophical debate between advocates of psychological egoism and those of psychological altruism. Defenders of psychological egoism maintain that human beings are intrinsically motivated by self-interest, such that every action, even one that appears altruistic, ultimately stems from an egoistic desire. Conversely, proponents of psychological altruism argue that there exist fundamental principles in human nature that lead individuals to act with the goal of promoting the well-being of others.

Defenders of psychological egoism generally advance a central argument: every altruistic action can be explained by a self-directed motivation, whether explicit or unconscious. In response, supporters of psychological altruism offer various counterarguments, among them that of Joseph Butler, stated in 1726. Butler established an essential link between desire and pleasure, asserting that desire must necessarily precede pleasure. For example, the pleasure of eating an apple can only exist if it was preceded by the desire to eat that apple. Therefore, pleasure cannot be the initial motivation of the action. This argument challenges the idea that pleasure, as a hedonistic pursuit, is the primary source of human motivation. In other words, if the desire for an external object or goal is independent of a hedonistic motivation, then it becomes impossible to explain all human actions as fundamentally egocentric or driven by the search for pleasure.

Defenders of psychological egoism respond by invoking the role of the unconscious in decision-making. According to them, our deep motivations often escape introspection and may be influenced by unconscious mechanisms that steer our actions toward egoistic goals.

Faced with this philosophical stalemate, it is relevant to turn to scientific advances to shed light on the debate. Research in neuroscience has explored the neural bases of altruism. Some studies reveal that specific brain areas associated with dopamine (often called the “pleasure hormone”) are strongly activated when individuals believe they are cooperating with other humans. Conversely, these same areas are significantly less active when participants learn that their interlocutor is a computer. These results seem, at first glance, to support the egoistic thesis, insofar as the neural pleasure associated with cooperation could explain altruistic behaviors.

However, defenders of psychological altruism raise an important objection: can we truly claim that the anticipation of this “dopaminergic reward” is the primary cause of cooperative behavior? This question brings the debate back to an impasse similar to that observed in the philosophical domain. Other studies, notably on the effects of punishment and revenge as sources of satisfaction, have not definitively refuted the thesis of psychological altruism.

In conclusion, and in accordance with the definition of psychological altruism, it appears that this debate remains unresolved. Philosophical and scientific approaches, although enriching, have not yet provided a definitive answer, leaving many avenues open to deepen our understanding of human nature.

2.4 Links between definitions

It has been seen so far that there are several ways to define altruism. To better understand the links between these definitions and their potential differences, it is useful to discuss and compare them.

First, the definitions of psychological and behavioral altruism are very similar but arise in different contexts. Behavioral altruism appears mainly in economic contexts where gains are monetary and measurable, whereas psychological altruism appears more in social science contexts, particularly psychology. In her book [1], Christine Clavier distinguishes these different forms of altruism, but many works consider these two definitions equivalent. Psychological altruism focuses on the intentions behind actions, while behavioral altruism analyzes situations in terms of costs and ben-

efits. In both cases, an altruistic action can be simply defined as a selfless action aimed at helping others.

However, there are certain specific examples of actions that do not simultaneously correspond to both definitions of altruism. For instance, in a situation where an agent has altruistic intent toward another but fails to actually help, from a psychological point of view the action is considered altruistic because the person had the genuine intention to help. In contrast, from a behavioral perspective, the action is not considered altruistic, since the other person received no effective benefit. In most cases, however, these two definitions coincide.

The similarities between biological and psychological altruism are less obvious. For example, giving poisoned food to a beggar without knowing it is toxic is considered psychological altruism, but not biological or behavioral altruism. Indeed, the deep motivations of the altruist, in this case, are to help the beggar, even though, biologically speaking, giving poisoned food does not promote the beggar's genes.

However, from a gene-centered perspective, there are many cases of altruism that are simultaneously psychological, behavioral, and biological. Indeed, from this perspective, it is possible to assert that when one genuinely intends to help someone and succeeds, it constitutes psychological, behavioral, and biological altruism—provided that the other person is genetically related.

To better understand the connection between the different types of altruism, it is relevant to take another perspective. Instead of starting from their respective definitions, we can formalize the contexts that give rise to altruistic actions and attempt to understand in which environment one type of altruism appears rather than another, or whether they appear simultaneously. To do so, we turn to game theory.

2.5 Link to game theory

Game theory arose from the need to understand the emergence of altruistic behaviors in nature, initially based on the gene-centered perspective, which considers genes as the true units of selection. According to this approach, only traits that favor gene transmission are selected, which explains the appearance of altruistic behaviors through kin selection. This mechanism, formalized by Hamilton's rule ($rb > c$), shows that individuals can sacrifice their own fitness to help relatives, thereby increasing the propagation of shared genes. However, kin selection alone cannot explain all forms of altruism, particularly among unrelated individuals. Reciprocal altruism thus broadens the cooperative framework, showing that repeated interactions can stabilize cooperation through reciprocity, provided that individuals can evaluate and remember past behaviors. To model these mechanisms, game theory is used, with games such as the prisoner's dilemma, which simulate decisions between cooperation and selfishness. Pioneering work such as the simulations of Robert Axelrod or Maynard Smith demonstrated how competing behavioral strategies evolve in given contexts, thereby mathematically explaining the emergence and stability of altruistic behaviors, even among initially selfish individuals [2]. Thus, fitness competition between agents allows modeling of biological altruism.

Behavioral altruism can also be modeled through scenarios where individuals make decisions involving costs to themselves and benefits to others, without expecting any return, immediate or delayed. Game theory allows the exploration of such interactions through the *Dictator Game* or the *Trust Game*, as mentioned in Subsection 2.2. In the dictator game, one individual (the "dictator") decides whether to share a sum of money with another player. Any decision to give up part of one's gains, even without expected benefit in return, is considered an altruistic behavior. This demonstrates that individuals, on average, do not always maximize their own profit, which aligns with the definition of behavioral altruism. The trust game, slightly different, involves interaction between two players. The first player decides to send a sum of money to a second player, called the recipient. The sum is multiplied, and the recipient can then choose to return part of this

sum to the first player. When participants take the risk of trusting without any guarantee that the other player will reciprocate, they exhibit altruistic behavior. Such behaviors can be studied through payoff matrices, where altruistic choices are less profitable in the short term but may prove advantageous in repeated interactions. This immediately recalls the *tit-for-tat* strategy in the prisoner’s dilemma, where the strategy consists of cooperating first and then exactly mirroring the opponent’s previous move [3]. Game theory thus also allows discussion of cases of behavioral altruism, even though these are more complex to simulate.

It is more difficult to find in the literature games that help understand and model the debate on psychological altruism. However, game theory could incorporate empathy or the desire for cooperation—internal and conscious motivations—by considering subjective preferences and cognitive dynamics. Classical models such as the prisoner’s dilemma assume that players maximize only their own payoff (final score). However, it would be interesting to introduce a social dimension, for example by integrating aversion to inequality, concern for others’ well-being, or the effect of emotions, which lies at the heart of the debate on psychological altruism. Thus, a player might accept a personal cost to reduce inequalities or improve collective well-being, reflecting a psychologically altruistic motivation. It is important to note that the more parameters are added, the less obvious the predictions become, since it becomes difficult to determine the causal influence of each. Thus, attempting to address this debate through game theory appears more complicated for this notion of altruism, even though a clear connection can still be drawn.

2.6 Related work

Our work falls within a rapidly expanding research field: the emergence of complex social behaviors in multi-agent environments. Two major contributions from the DeepMind team help situate our approach and justify our inspirations, while highlighting paths we could have explored or developed further.

In their article *Multi-agent Reinforcement Learning in Sequential Social Dilemmas* [4], Leibo et al. challenge the classical modeling of social dilemmas using matrix games (such as the prisoner’s dilemma), showing that these do not capture the complexity of sequential interactions in the real world. Instead, they introduce the notion of the Sequential Social Dilemma (SSD), where cooperative or defecting behaviors are not simple actions but entire policies, that is, long-term strategies learned through experience.

This distinction could have enriched our own approach, particularly by offering a finer interpretation of the autonomous agent’s actions. For instance, in our simulations, an agent that gives fitness to another can be seen as “cooperative,” but this isolated action does not necessarily constitute a long-term cooperative policy. By integrating a policy-based analysis as in Leibo et al., we could have better characterized the social nature of emerging behaviors and more rigorously distinguished opportunistic cooperative tendencies from genuinely altruistic strategies.

Moreover, their empirical analysis of payoff matrices derived from opposing policies (cooperation vs. defection) could have provided a systematic evaluation framework to classify emerging dynamics in our grid environment, similar to their distinction between “Gathering” and “Wolfpack” games.

The article *Emergent Bartering Behaviour in Multi-Agent Reinforcement Learning* [5] explores how, without explicit instructions, agents can learn to exchange resources when trained in a sufficiently expressive environment. This result highlights the ability of multi-agent environments to generate spontaneous forms of social organization, as long as interaction mechanisms are rich and sufficiently incentivizing. This work could have inspired us in several ways.

First, the richness of the environment implemented in their study is very comprehensive. Their simulation integrates distinct resources, differentiated preferences, and a more diverse action space. Introducing such complexity into our own model—for example, by adding the notion of resources or exchange mechanics—could have made the emergence of altruistic behaviors more meaningful, incorporating more realistic forms of giving or bartering. However, as will be clarified later, the

goal is not to simulate a realistic situation but rather to analyze the emergence of altruistic strategies within various populations.

Next, the notion of instrumental cooperation is very interesting. In their study, bartering emerges as a strategic compromise without moral motivation. This could have fueled our reflection on behavioral altruism, which, as explained earlier, is not straightforward to link to game theory or computer simulations.

Finally, simulating a social economy introduces a new dimension and opens the way toward a more economic rather than biological or psychological modeling of altruism, depicting agents that exchange rather than merely give or attack. This could expand our framework toward simulating the foundations of economic cooperation.

3 Methodology: Computer simulation of altruism

3.1 Presentation of the simulation model

3.1.1 Definition of the environment

The environment of our simulation takes the form of a two-dimensional grid of 20x20 cells, representing a closed space where several distinct populations interact, described precisely later. In Fig.(1), three distinct populations are shown in red, green, and blue.

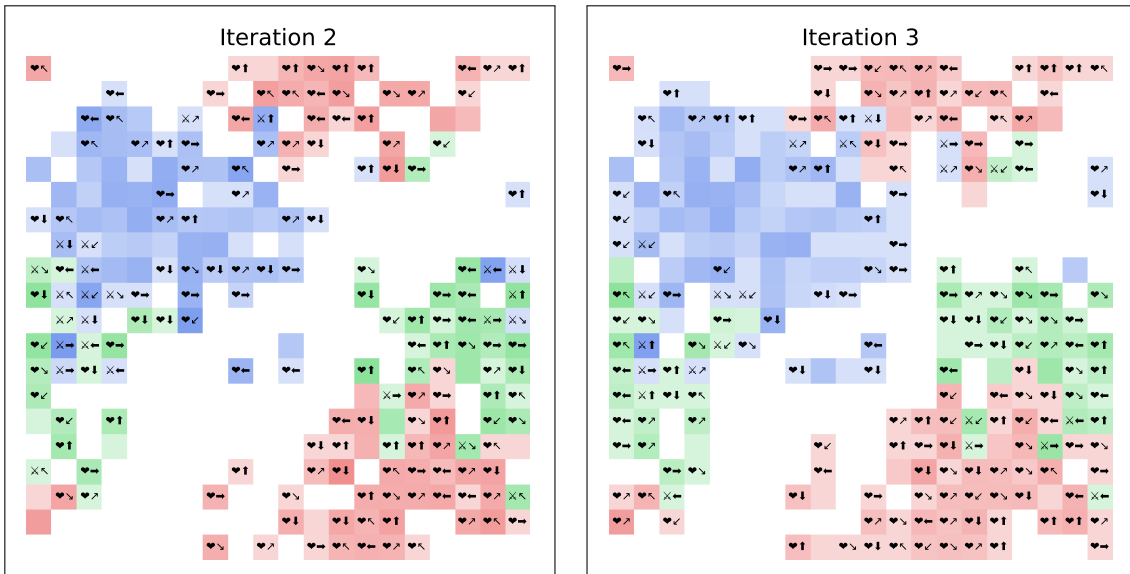


Figure 1: Simulation environment at the 2nd then the 3rd iteration with “giving” agents in red, “attacking” in blue and “complete” in green on a 20x20 grid.

Each cell can potentially be occupied by an agent, and the boundary conditions are periodic, that is, actions passing through one edge act on the opposite side of the grid. This makes it possible to remove potential edge effects that could distort the emergent behaviors of the different agents. As can be seen in Fig.(1), the action performed by each agent during the last iteration is visible. The bold arrows represent the direction of actions, and the hearts and swords represent donations and attacks, respectively. If no symbol appears on a cell, this means that the agent decided to do nothing. White cells represent unoccupied cells, and the colored cells have different color intensities; the darker the cell, the more fitness the agent has. Below a certain fitness threshold, the agent disappears and the cell becomes white.

Populations are initialized according to Gaussian distributions centered randomly on the grid, which favors the natural formation of clusters. Within these clusters, the fitness levels of agents (their selective value) also follow a normal distribution, which introduces the notion of heterogeneity within the same population, thereby making the simulation more realistic. To do this, the mean fitness was set to a value of 1, with a standard deviation of 0.2.

Agents have three possibilities at each new iteration. They can do nothing, attack, or give. In the case of a donation, 20% of the agent's fitness is removed and is distributed to the agent in the chosen direction. Moreover, agents can choose to attack a neighboring cell. In this case, the smaller fitness between the agents in the duel is subtracted from the fitness of both agents. This implies that the agent with the smaller fitness dies, and the other takes damage equal to the eliminated agent. Finally, the original fitness of the killed agent is redistributed equally among the neighboring cells. For example, if agent A with a fitness of 0.5 attacks agent B who has a fitness of 0.8, then agent A dies and agent B ends up with 0.3 fitness. Then, agent A's fitness before the attack (0.5) is redistributed equally to the neighboring cells.

There are also two other parameters that we varied during our simulations; the donation bonus and the growth. Indeed, these two parameters have a great influence on the outcome of the results. Thus, the environment is not fixed and the conditions for the emergence of altruistic behaviors can be analyzed more precisely.

The donation bonus is a percentage bonus that the cell receiving a donation will obtain. For example, if this bonus is 4%, then a cell will receive 24% of the donor's fitness, but the latter will lose only 20%. This parameter serves to model real cases of altruism where the altruistic action often has much less negative effect on the donor than positive effect on the recipient. This parameter was varied between 0 and 8% (0%, 4% and 8%).

One problem we observed is that the amount of available fitness is limited and fixed at initialization (at least without a donation bonus). As a result, new cells are increasingly weak and struggle to repopulate the entire initial grid. To address this problem we defined the growth parameter. The latter is an absolute amount of fitness that all cells on the grid receive at each iteration. This parameter can be seen as the growth of each individual in the environment; they become a little stronger at each new iteration. This makes it possible to counter the problem of individuals becoming increasingly weak. This parameter was varied between 0 and 0.08 (0, 0.04 and 0.08), and represents an absolute amount of fitness.

At the start, the cost to occupy a new cell was zero. Subsequently, as detailed in Section 4.1, this was modified so that an agent must make a donation on an empty cell to create a new agent and spread the population across the grid.

3.1.2 Introduction of agents

Three types of agents were designed to explore different levels of behavioral complexity:

- **Completely random agents:** These agents act without distinguishing ally from enemy, choosing their actions (attack, donation, inaction) randomly from all possibilities. Unable to make strategic choices, they serve as a baseline for more strategic behaviors.
- **Semi-random agents:** These agents follow simple rules imitating a partially strategic behavior, or "semi-random." They attack only agents from other populations or reserve their "fitness" gifts exclusively for their allies, thereby reinforcing cooperation within the same population. There are three types of semi-random agents; those that only make donations to their allies (*Giving*), those that only attack enemies (*Attacking*), and the *Complete*, who either give to their allies or attack their enemies at random.

- **Intelligent agent:** This type of agent forms the basis of our investigation, using a reinforcement learning algorithm (Reinforce [6]) to maximize its cumulative “fitness” over time. Rewards are assigned based on changes in fitness in the occupied cells, and penalties are imposed if the population to which it belongs disappears, thereby encouraging agents to converge toward an optimal strategy. The definition of this convergence is explained later.

3.2 Simulation parameters and learning of the autonomous agent

To simulate behaviors close to altruism, we designed a system in which agents learn to interact with their environment strategically. The main agent, which we call “intelligent,” operates using a reinforcement learning algorithm [6], based on the principle of reward and punishment. Concretely, this agent performs actions such as moving, attacking another agent, or giving part of its own “fitness,” and receives a reward or a penalty depending on the consequences of its choices. The goal of this agent is to maximize its long-term reward. For example, if it gives part of its fitness to an allied agent and this strengthens its population over subsequent iterations, it can benefit indirectly in the future and learn to take the right action in a given situation already encountered. Conversely, if it acts in a disorderly manner and weakens its own population, it will be penalized and will learn from it. The autonomous agent therefore learns via a neural network, trained by trial and error: at each simulation, it tests actions, observes the results, and adjusts its future decisions accordingly. The autonomous agent’s network is composed of two linear layers, separated by ReLU activation functions and dropout regularization. This process is framed by several parameters, described below:

- Learning rate controlling the speed at which the agent adjusts its strategies: 0.001.
- Discount factor that values the rewards the agent expects to obtain in the near future rather than the distant future: 0.98.
- Batch size: 50 (the agent learns from 50 experiences simultaneously).
- Grid size: 20x20 cells.

The agent’s choices (attack, give, do nothing) are decided from several possible options, and their probability is adjusted by the algorithm over time. A method called “entropy regularization” is used to encourage the agent not to lock too quickly onto a single strategy, continuing to explore other possibilities during training. This makes it possible to converge toward a strategy after having analyzed several possibilities, and thus avoid immediate convergence. The Reinforce algorithm is used to find an optimal strategy. This algorithm is simple to implement, but is known to be difficult to make converge. In our trials, we noticed that adding entropy regularization allowed Reinforce to converge more easily, or in other words, to find a solution more often and more quickly. In addition, we added certain elements of the “Proximal Policy Optimization” (PPO) algorithm [7], such as “Gradient Weight Clipping,” to facilitate the convergence of learning.

The convergence of learning is observed through two main indicators: the stabilization of cumulative rewards over episodes, and the progressive decrease of the “policy loss,” which is what we seek to minimize during the agent’s training. When these two measures reach a stable plateau, we consider that the agent has acquired a coherent strategy in the given environment.

3.3 Assumptions and limitations of the model

Biological altruism refers to behaviors that, although costly to the individual, increase the chances of survival of its genetic lineage or group of belonging. This type of altruism can be modeled in our simulation: semi-random agents give fitness only to their similar ones, which favors collective survival. Likewise, the autonomous agent can learn to strengthen agents of its own

group if this indirectly improves its own performance. Thus, a functional equivalent of biological altruism emerges within the simulation framework, even though it does not rely on actual genetic transmission.

Behavioral altruism is manifested by actions that harm the individual without any expected benefit in return. Yet, our autonomous agent is trained to maximize its cumulative reward: every action taken aims, explicitly or implicitly, to increase its own “fitness.” It is therefore not possible, in our framework, to observe a behavior that would be purely altruistic in the strict behavioral sense, that is, detached from any form of self-interest, even deferred. The question arises for semi-random agents, where agents make decisions without expecting a benefit in return. Indeed, when attacking an agent from another population, it does not know whether its “sacrifice” of fitness will benefit its population or itself, since it has not been trained to perform this action.

As explained previously (Section 2.5), it is complicated to link psychological altruism to game theory, or in our case, to computer simulations. Psychological altruism implies an internal intention to help others, motivated by emotions such as desire or empathy. This type of altruism presupposes a form of self- and other-awareness that is absent in our simulation framework. Our agents are neither conscious nor empathic; their actions are entirely guided by utility calculations. Even when an agent acts in favor of another, this results from a strategy aiming at a future gain, and not from a sincere concern for others’ well-being. Consequently, our simulations cannot account for psychological altruism, which relies on subjective mental states that our agents do not possess.

Beyond the conceptual limits related to the types of altruism, our model relies on technical assumptions that influence the scope and validity of the results.

The simulated universe is a two-dimensional grid of 20x20 cells. This is an extreme simplification of the real world, where agents interact only with their immediate environment (their direct neighbors), there are neither natural resources, nor dynamic changes in the terrain, nor uncertainty other than that integrated into their decisions. This simplification is necessary to make the problem technically solvable, but it limits the complexity of strategies and the diversity of possible behaviors. For example, complex cooperation, reputation, or cultural transmission are absent. The goal, however, is not to approach reality and to show the potential emergence of altruism, because in that case, one would rather observe and discuss cases encountered in nature. Here, the simulation aims to observe whether biological or behavioral altruistic behaviors can emerge when facing different types of simplified populations, and if so, the strategies adopted.

Agents are initially distributed according to Gaussian distributions centered randomly on the grid, and their fitness level varies according to another normal distribution. This method favors the formation of groups and simulates a certain heterogeneity. However, it also imposes a stable structure from the start, which could artificially favor cooperation within a group. One might ask whether a more chaotic distribution would change the optimal strategies acquired by the autonomous agents.

The discount factor plays a fundamental role in the learning logic. It influences future rewards, because the closer it is to 1, the more the agent values long-term benefits, and the lower it is, the more it favors immediate gains. The choice of 0.98 reflects a desire to favor long-term strategies. Thus, actions that bring a benefit only in the short term are automatically disadvantaged. This confirms what was said earlier, namely that behavioral altruism cannot be observed for autonomous agents within this simulation, because a short-term benefit is the very definition of behavioral altruism, also called “selfless altruism.” By varying this parameter, this would help preserve truly altruistic behaviors in the strict sense, which offer no return on investment, even deferred, but it would increase the probability of immediate convergence without exploration of all possibilities. Yet, within the framework of this work, the emerging altruistic strategy is sought.

Finally, the Reinforce algorithm used is a classic reinforcement learning algorithm, which assumes that rewards are available and measurable at each step. In addition, this algorithm requires the Markov assumption [8], which posits that the future state depends only on the current state

and present action, and not on the past history. The first assumption is not neutral, because it prevents agents from taking an action whose utility would be moral or symbolic. Moreover, as stated earlier, psychological altruism is therefore not represented, because internal intentions are not motivated by emotions, but by quantifiable calculations.

4 Results

4.1 Problems encountered

When obtaining the first results, a problem arose during the duels of the intelligent agents against the semi-random ones, but also for the scenario detailed below. As explained previously, in a first version, it cost nothing for an agent to populate a new cell. Thus, the behavior of the intelligent agents is as follows: occupy a cell, wait for a neighboring cell to be killed to receive a portion of the redistributed fitness, and repopulate the cell for free. The results of this situation can be visualized in Fig.(3a).

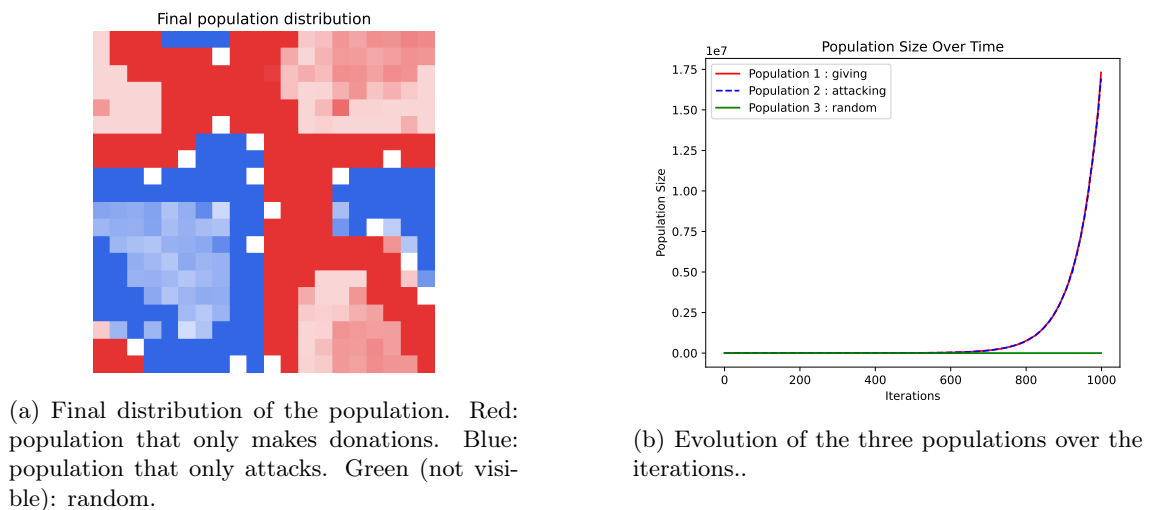


Figure 2: Results of the battle between donating populations (red), attacking populations (blue), and random populations (green).

As can be observed in Fig.(3a), this problem leads to empty cells surrounded by agents possessing high fitness. By waiting in the second row behind a battle, the populations accumulate fitness, which grows exponentially, as shown in Fig.(3b). The random population, which does not establish any strategy, does not develop at all. This behavior was also observed among the intelligent agents, meaning they do not perform as expected in this environment. Assigning a cost to occupying a cell solved this issue and allowed us to obtain the results presented in the following section.

4.2 Intelligent agent vs semi-random

The autonomous agent, called the “intelligent agent,” fought against two types of semi-random agents: the giving and attacking agents. To observe the evolution of the intelligent agents’ strategies, the donation bonus and growth parameters were varied, as explained in the methodology of this work.

4.2.1 Intelligent agent vs giving

To begin, a population of intelligent agents fought against two populations of semi-random agents that only perform donations, called “giving.” The results obtained, by varying the donation and growth bonuses, are shown in the following table:

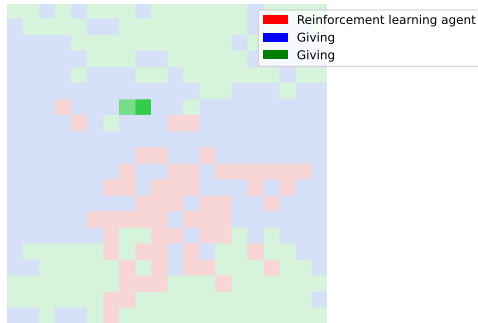
Table 1: Distribution of the fitness of the different populations in % and total fitness as a function of the donation and growth bonuses for a population of intelligent agents against two populations of semi-random “giving” agents, with in bold the population having the highest fitness at the last iteration of the simulation.

	Population	Donation bonus [%]/growth [-]								
		0/0	0/0,04	4/0	4/0,04	8/0	8/0,04	0/0,08	4/0,08	8/0,08
Fitness[%]	Intelligent agent	42,23	67,41	36,93	21,66	7,12	3,39	73,26	16,06	2,71
	Giving	28,97	16,91	28,13	41,12	40,70	42,08	13,79	45,59	43,93
	Giving	28,80	15,68	34,94	37,22	52,18	54,53	12,95	38,35	53,36
Total Fitness		104,50	4471,60	466,36	31867,81	3,43e12	1,17e14	6610,87	58351,46	2,76e14

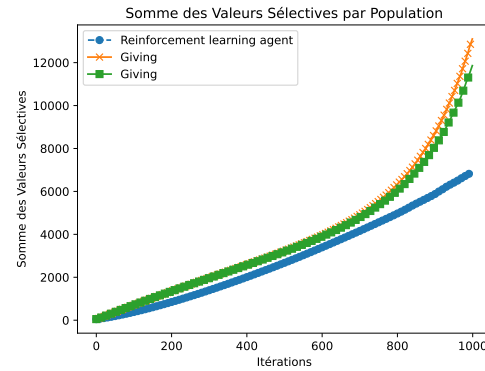
The intelligent agents manage to find a winning strategy against the giving agents as long as the donation bonus remains low. The giving agents also seem to be slightly favored by a growth bonus applied at each iteration.

The final distribution of the populations, as well as the evolution of the three populations over the iterations, can be visualized for each simulation with specific donation and growth bonuses. For example, for a donation bonus of 4% and a growth bonus of 0.04, the following results were obtained:

Exemple de Distribution Finale des Populations



(a) Final distribution of the population. Red: intelligent agents. Blue and green: semi-random “giving” agents.



(b) Evolution of the three populations over the iterations.

Figure 3: Results of the battle: intelligent agent vs giving vs giving.

The distributions and graphs are not presented for each donation/growth bonus ratio to avoid an overload of information, but they will be analyzed and discussed later for those that provide additional insights.

4.2.2 Intelligent agent vs attacking

The intelligent agents win only when the growth bonus is zero. As soon as it becomes nonzero, the intelligent agents fail to find a winning strategy against the attacking agents.

Table 2: Distribution of the fitness of the different populations in % and total fitness as a function of the donation and growth bonuses for a population of intelligent agents against two populations of attacking agents.

	Population	Donation bonus [%]/growth [-]								
		0/0	0/0,04	4/0	4/0,04	8/0	8/0,04	0/0,08	4/0,08	8/0,08
Fitness[%]	Intelligent agent	34,69	5,57	34,28	5,11	33,84	4,96	2,32	2,42	2,12
	Attacking	32,51	49,91	32,70	46,76	32,92	47,73	47,05	49,17	44,90
	Attacking	32,80	44,52	33,02	48,12	33,23	47,31	50,63	48,42	52,98
Total Fitness		124,17	10746,65	152,84	10834,52	206,60	10670,33	20692,76	20656,42	20718,66

4.3 Intelligent agent vs random

Regardless of the donation/growth bonus ratio, the intelligent agents outperform the random agents. However, the random agents perform slightly better as soon as a growth bonus is introduced.

Table 3: Distribution of the fitness of the different populations in % and total fitness as a function of the donation and growth bonuses for a population of intelligent agents against two populations of random agents, with in bold the population having the highest fitness at the last iteration of the simulation.

	Population	Donation bonus [%]/growth [-]								
		0/0	0/0,04	4/0	4/0,04	8/0	8/0,04	0/0,08	4/0,08	8/0,08
Fitness[%]	Intelligent agent	40,48	35,75	40,18	35,64	40,10	35,56	35,68	35,44	35,48
	Random	30,61	35,05	30,78	35,05	31,01	35,00	35,06	34,96	34,95
	Random	28,92	29,20	29,04	29,32	28,89	29,44	29,27	29,60	29,57
Total Fitness		88,35	6197,86	97,08	6260,44	109,98	6275,02	10771,16	10978,21	11159,79

4.4 Semi-random vs random

Each type of semi-random agent fought against the random agents in order to observe their behaviors and strategies.

4.4.1 Giving vs random

The giving agents always win against the random agents. However, when the growth bonus is zero, all agents on the grid quickly weaken and disappear.

Table 4: Distribution of the fitness of the different populations in % and total fitness as a function of the donation and growth bonuses for a population of semi-random agents that only make donations (“giving”) against two populations of random agents, with in bold the population having the highest fitness at the last iteration of the simulation.

	Population	Donation bonus [%]/growth [-]								
		0/0	0/0,04	4/0	4/0,04	8/0	8/0,04	0/0,08	4/0,08	8/0,08
Fitness[%]	Giving	0	41,11	0	41,17	100	41,26	41,09	41,16	41,28
	Random	0	29,44	0	29,42	0	29,37	29,46	29,42	29,36
	Random	0	29,44	0	29,42	0	29,37	29,46	29,42	29,36
Total Fitness		0	181,18	0	206,02	1,60e6	239,13	361,00	411,46	479,07

4.4.2 Attacking vs random

The same pattern is observed here. The attacking agents consistently outperform the random agents as soon as the growth bonus is nonzero (leading to the disappearance of all agents when it is zero).

Table 5: Distribution of the fitness of the different populations in % and total fitness as a function of the donation and growth bonuses for a population of semi-random agents that only attack (“attacking”) against two populations of random agents, with in bold the population having the highest fitness at the last iteration of the simulation.

	Population	Donation bonus [%]/growth [-]								
		0/0	0/0,04	4/0	4/0,04	8/0	8/0,04	0/0,08	4/0,08	8/0,08
Fitness[%]	Attacking	100	41,64	100	41,85	100	42,14	41,67	41,87	42,12
	Random	0	29,18	0	29,07	0	28,93	29,16	29,07	28,94
	Random	0	29,18	0	29,07	0	28,93	29,16	29,07	28,94
Total Fitness		7,40e-07	187,89	2,48e-05	207,05	6,10e-04	231,29	375,11	413,72	463,48

4.4.3 Complete vs random

Once again, the semi-random agents outperform the random agents, regardless of the donation and growth parameters.

Table 6: Distribution of the fitness of the different populations in % and total fitness as a function of the donation and growth bonuses for a population of complete semi-random agents (giving + attacking) against two populations of random agents, with in bold the population having the highest fitness at the last iteration of the simulation.

	Population	Donation bonus [%]/growth [-]									
		0/0	0/0,04	4/0	4/0,04	8/0	8/0,04	0/0,08	4/0,08	8/0,08	
Fitness[%]	Compleat	0	41,08	0	41,15	100	41,24	41,06	41,15	41,25	
	Random	0	29,46	0	29,43	0	29,38	29,47	29,42	29,38	
	Random	0	29,46	0	29,43	0	29,38	29,47	29,42	29,38	
Total Fitness		0	181,03	0	206,09	1,46e06	238,32	362,54	411,18	478,41	

4.5 Semi-random vs semi-random

After establishing a baseline comparison of the semi-random agents’ behaviors by having them compete against non-strategic agents, the three types of semi-random agents were then pitted against each other.

4.5.1 Complete vs giving vs attacking

When the donation bonus is high, the giving agents win. In other situations, the attacking agents are stronger. However, it was observed that the results depend on how the populations are initialized on the grid. Indeed, in some cases, the complete agents won, although this is not reflected in the results shown in the following table:

Table 7: Distribution of the fitness of the different populations in % and total fitness as a function of the donation and growth bonuses for the three different populations of semi-random agents (complete, giving, attacking), with in bold the population having the highest fitness at the last iteration of the simulation.

	Population	Donation bonus [%]/growth [-]								
		0/0	0/0,04	4/0	4/0,04	8/0	8/0,04	0/0,08	4/0,08	8/0,08
Fitness[%]	Completing	0	0	0,34	0,88	34,70	32,71	0	0,60	34,27
	Giving	0	0	0,39	0,80	65,30	67,29	0	0,49	65,73
	Attacking	100	100	99,27	98,32	0	0	100	98,91	0
Total Fitness		26,84	1,49e04	18,56	1,21e04	4,13e12	1,74e13	2,98e04	2,50e04	3,17e13

5 Discussion: Relevance and implications of the results

5.1 Analysis of the results

5.1.1 Varied parameters

The results show a strong influence of the “donation bonus” and “growth bonus” parameters on the performance of the different strategies, as discussed in more detail in the following sections. The attacking agents spread rapidly and benefit greatly from growth, while the giving (and complete) agents benefit from the donation bonus. However, it is interesting to observe that these two parameters appear to be independent: in simulations where the growth bonus is nonzero, the attacking agents almost always win, regardless of their opponents or the donation bonus. Conversely, in simulations with a high donation bonus (greater than or equal to 8%), the giving agents win regardless of the growth value.

In reality, these two parameters are not completely independent. Their correlation effect can be observed in the “intelligent agent vs giving” simulation (Table 1). In the environment with a 4% donation bonus and no growth, the intelligent agent wins, whereas in the simulation with a 4% donation bonus and 0.04 growth, the giving agent wins. It seems that the giving agent benefits from growth to strengthen its donations, allowing it to win the simulation. This weak correlation can be explained simply by the fact that growth increases the fitness of all individuals, thereby slightly increasing the absolute amount of fitness donated (since it is a percentage of the donor’s fitness). However, because the fitness of all individuals increases, this slight bonus effect is often very small and therefore difficult to observe.

5.1.2 Intelligent agent vs semi-random

The intelligent agents first fought against agents that only make donations. In this configuration, two distinct populations of “giving” agents face one population of intelligent agents. The giving agents only donate fitness to their own members, never attacking or helping other populations. Their non-strategic internal generosity creates a situation favorable to strategic parasitism: the RL agent benefits from the redistribution of fitness among the giving agents, which weakens them. The intelligent agents, by donating more strategically, eliminate the giving agents, who have higher fitness costs. However, when the donation bonus is high ($\geq 4\%$), the giving agents outperform the intelligent ones, regardless of the growth bonus. This is explained by an effective cooperative dynamic among the giving agents, where donations circulate internally, mutually reinforcing their fitness. The intelligent agents, though adaptive, cannot compete with this efficient solidarity in a generous environment. The intelligent agent manages to adopt an interesting strategy when the growth bonus is zero:

Exemple de Distribution Finale des Populations

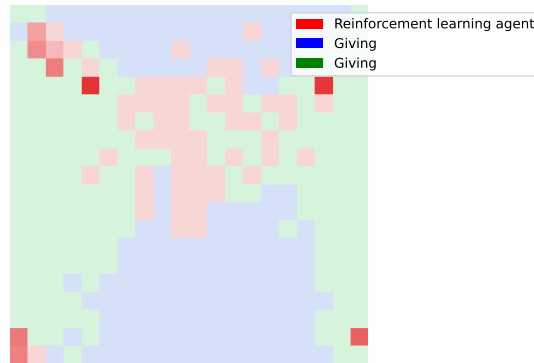


Figure 4: Final population distribution during the battle of the intelligent agent vs giving, for a donation/growth bonus of 0%/0.04.

It can be seen that the intelligent agents do not necessarily occupy a large portion of the environment but still gather most of the total fitness (67.41%), thanks to a few individuals holding very high fitness. These agents accumulate fitness and benefit from the self-destruction of the giving populations that weaken each other through repeated donations without collective return. They simply collect redistributed fitness without taking risks, eventually becoming “unbeatable.” For this reason, darker red cells appear in Fig.(4).

The “attacking” agents only target members of other populations, never their own allies. They thus form two coordinated populations around a simple offensive-defensive goal: weakening the other groups. The intelligent agent only finds a winning strategy when the growth bonus is zero. In this case, the attacking agents, by attacking each other predictably, often self-destruct, creating opportunities for the intelligent agents to expand, regardless of environmental bonuses. Interestingly, the intelligent agents initially lose fitness faster than the giving ones, even when the growth bonus is zero:

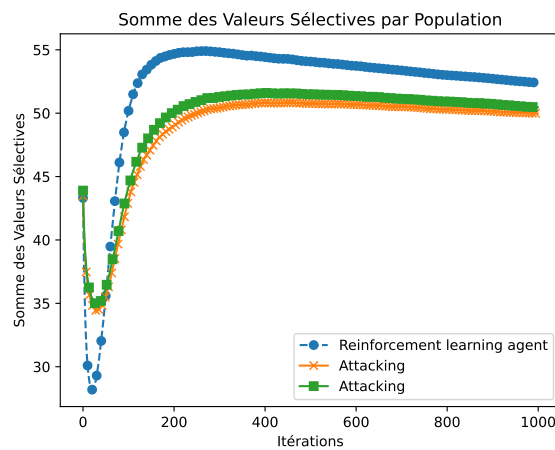


Figure 5: Sum of fitness values per population for intelligent agent vs attacking, for a donation/growth bonus of 4%/0.

Their strategy appears to involve spreading quickly at first to occupy a large portion of the grid and collect fitness later on. At the beginning of the simulation, the attacking agents lose less fitness because they attack slightly weakened intelligent agents, but they soon become outnumbered. Interestingly, the intelligent agents adopt different strategies depending on the opponent: against

giving agents, a few individuals accumulate large amounts of fitness and remain stationary; against attacking agents, they exploit numerical superiority to overwhelm the enemy. However, when the growth bonus is nonzero, the attacking agents can quickly regain lost fitness and maintain offensive pressure. In these cases, the intelligent agents do not systematically dominate, especially if their learning does not lead them to avoid attacks efficiently. They seem unable to flee or react effectively, being eliminated even when they learn good positioning or energy conservation. Each attack reduces the fitness of both attacker and target, but the coordination among attackers allows them to absorb the cost more efficiently by surrounding and eliminating the intelligent agents. Even when trained, a population of intelligent agents cannot survive against two coordinated, offensive enemy populations in an environment where agents receive fitness each iteration. It is also important to note that the intelligent agents were trained in a zero-growth environment. It would be interesting to see whether their strategy would evolve if trained under different conditions, which is very likely.

When facing random agents, the intelligent agents clearly dominate in every case. This confirms that they have been properly trained and effectively develop a strategy. Random agents, having no strategy, serve as a useful baseline for comparison with the more complex agents. Random agents lack stable strategies, allowing intelligent agents to learn and exploit the most advantageous configurations. This dominance is slightly reduced when the growth bonus becomes positive, as random agents are “forgiven” for poor decisions and survive longer, though without changing the outcome.

It is interesting to note that there is no link between the donation and growth bonuses. Beyond a certain donation bonus, the intelligent agents can no longer beat the giving agents, just as they cannot beat the attacking agents once any growth bonus is introduced, regardless of the other parameter. Therefore, there is no fixed donation/growth ratio from which a specific winning strategy emerges.

5.1.3 Semi-random vs random

When the “giving” population faces two random ones, the outcome depends on the growth bonus. As soon as it is nonzero, the giving population wins by a wide margin. When a growth bonus is introduced, the semi-random agents have, on average, higher fitness because they make more “strategic” donations without depleting themselves too quickly. As the number of donations decreases, fitness losses are reduced and the giving population wins. However, when the growth bonus is zero, all three populations collapse immediately because both types of agents weaken without receiving any fitness between iterations. This explains the total fitness of zero in the 0%/0 and 4%/0 cases in Table 4.

Unsurprisingly, the attacking population completely dominates the random ones. They focus their attacks on enemies, while the random agents lack any strategy. A targeted offensive strategy and internal cohesion consistently outperform randomness. As before, this dominance is slightly reduced when a growth bonus is introduced, allowing the random agents to survive slightly longer.

The “complete” semi-random agents alternate between giving and attacking. Their performance depends, like that of the attacking and giving agents, on the growth bonus. As soon as this parameter is nonzero, the complete agents win, as observed with the giving ones. When the growth bonus is zero, their excessive generosity weakens them, as they prioritize giving over attacking, greatly reducing their fitness. The results are very similar to those observed with the giving agents.

Overall, the different types of semi-random agents achieve similar results against random agents. Using random agents as a non-strategic baseline is useful for assessing whether strategic behavior develops in the opponent, but it does not allow for deeper discussion.

5.1.4 Semi-random vs semi-random

This scenario highlights the limits and complementarities of the three semi-random strategies. In a “giving vs attacking vs complete” scenario, the attacking population wins in most cases. When the donation bonus is high (0.8), the giving agents can still succeed by forming internal cooperation networks that outweigh the sporadic attacks of others. If they are not too isolated, they manage to maintain a stable or even dominant population. The attacking agents thrive in environments with low growth and donation bonuses, where their targets struggle to recover lost fitness. The complete agents do not emerge in these conditions, which favor either attacking or giving populations. An intermediate environment with less extreme bonus values might encourage their emergence.

6 Conclusion

This work explored the conditions for the emergence of altruism through an original computational model based on agents evolving in a discrete environment. By pitting populations of intelligent agents trained with an RL algorithm against semi-random and random agents, and by varying key environmental parameters such as the donation bonus and the growth bonus, it was possible to test the impact of different strategies on the survival, spread, and collective success of the populations.

The results show that altruism can emerge in specific contexts, but that this emergence depends strongly on the structural conditions of the environment. For example, the giving agents prevail only when the system amplifies the benefits of giving (high donation bonus), while the intelligent agents adapt their behavior to the environment and often exploit systematic altruists. Paradoxically, their victory is not always synonymous with a truly altruistic strategy: against attacking agents, their behavior tends toward cooperation and diffusion, whereas against giving agents, they favor fitness concentration. This contrast suggests that effective altruism is not necessarily that which sacrifices, but that which balances giving and preservation.

The analysis also showed that the observed behaviors, though altruistic in appearance, do not all fall under the same type of altruism. The semi-random agents have neither intentionality nor adaptability: their behavior approaches behavioral altruism, or even a simple determinism without cognitive depth. They cannot in any case embody psychological altruism, nor can the intelligent agents, since they lack internal motivation. This highlights a tension between simulated altruism and emergent altruism, which is closer to natural evolutionary processes.

Finally, this project raises a broader question: what are the limits of simulations in modeling social behavior? While algorithms make it possible to test precise hypotheses and reveal certain non-trivial dynamics, they remain abstractions that neglect the cognitive, emotional, and historical richness of human or animal behaviors. Altruism, as a biological, social, and psychological phenomenon, cannot be reduced to a single metric or strategy. Moreover, more complex algorithms would make it possible to train the intelligent agents better, challenging the limits of modeling social behaviors and the study of emergent strategies.

This work is intended above all as an experimental basis for the study of the emergence of altruistic strategies in simulated environments. It sets simple but extensible foundations, showing how basic rules and localized interactions can lead to complex collective dynamics. Nevertheless, it is clear that the algorithms used here can be further developed, particularly on the side of the intelligent agents, to explore more subtle, adaptive, and contextual behaviors. Likewise, the parameters fixed in the simulations can be varied or refined to test other ecological or social conditions. The goal is not to freeze a model, but to open a modular framework that allows, step by step, a deeper computational understanding of altruism in its multiple forms.

References

- [1] CLAVIEN Christine. *Je t'aide...moi non plus*. Philosophie des sciences. Vuibert, 2010. ISBN: ISBN 978-2-311-00049-8.
- [2] Bruno BEAUFILS. "Modèles et simulations informatiques des problèmes de coopération entre agents". In: *Doctoral Thesis* (2009), pp. 1–6. DOI: <https://theses.hal.science/tel-00366446v1>.
- [3] AXELROD. *The evolution of cooperation*. Basic Books, 1984.
- [4] Leibo Joel et al. "Multi-agent Reinforcement Learning in Sequential Social Dilemmas". In: *Proceedings of the 16th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2017)* (2017), pp. 464–473. DOI: 10.5555/3091125.3091194.
- [5] Johanson Mike, Hughes Edward, and Leibo Joel. *Emergent Bartering Behaviour in Multi-Agent Reinforcement Learning*. URL: <https://deepmind.google/discover/blog/emergent-bartering-behaviour-in-multi-agent-reinforcement-learning/>. (accessed:23.04.2025).
- [6] Ronald J. Williams. "Simple statistical gradient-following algorithms for connectionist reinforcement learning". en. In: *Machine Learning* 8.3 (May 1992), pp. 229–256. ISSN: 1573-0565. DOI: 10.1007/BF00992696. URL: <https://doi.org/10.1007/BF00992696> (visited on 04/28/2025).
- [7] John Schulman et al. *Proximal Policy Optimization Algorithms*. arXiv:1707.06347 [cs]. Aug. 2017. DOI: 10.48550/arXiv.1707.06347. URL: <http://arxiv.org/abs/1707.06347> (visited on 04/28/2025).
- [8] "Generators and Markov Processes". en. In: *Markov Processes*. John Wiley & Sons, Ltd, 1986, pp. 155–274. ISBN: 978-0-470-31665-8. DOI: 10.1002/9780470316658.ch4. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470316658.ch4> (visited on 04/28/2025).